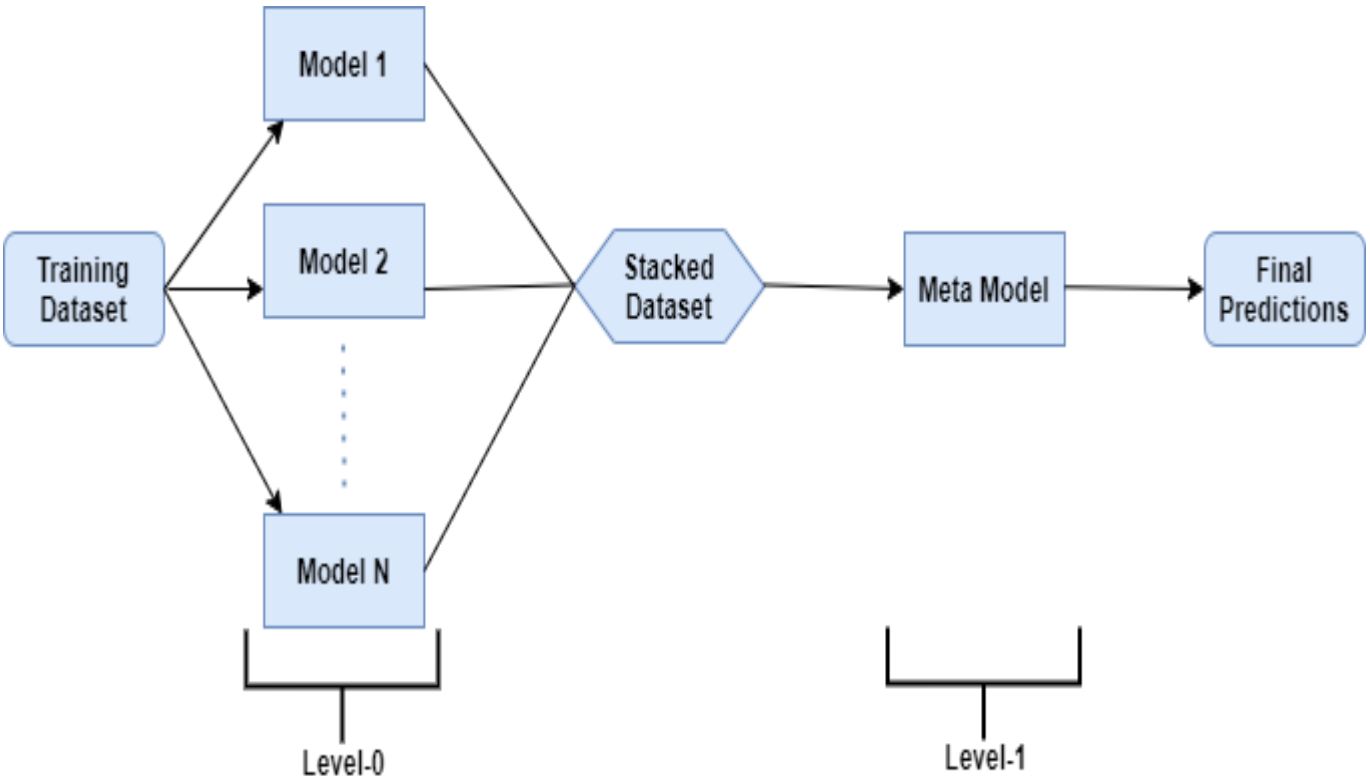
Ensemble Learning - Stacking

Stacking, also known as stacked generalization, is an ensemble learning technique that combines multiple machine learning algorithms via meta learning (either a meta-classifier or a meta-regressor).

The base-level algorithms are trained on the entire training dataset, and then the meta-model is trained on the predictions from all the base-level models as features. The base models are called level-0 models, and the meta-model which combines the base model's predictions is called a level-1 model.



The level-1 model training data is prepared via k-fold cross-validation of the base models, and out-of-fold predictions (real numbers for regression and class labels for classification) are used as the training dataset.

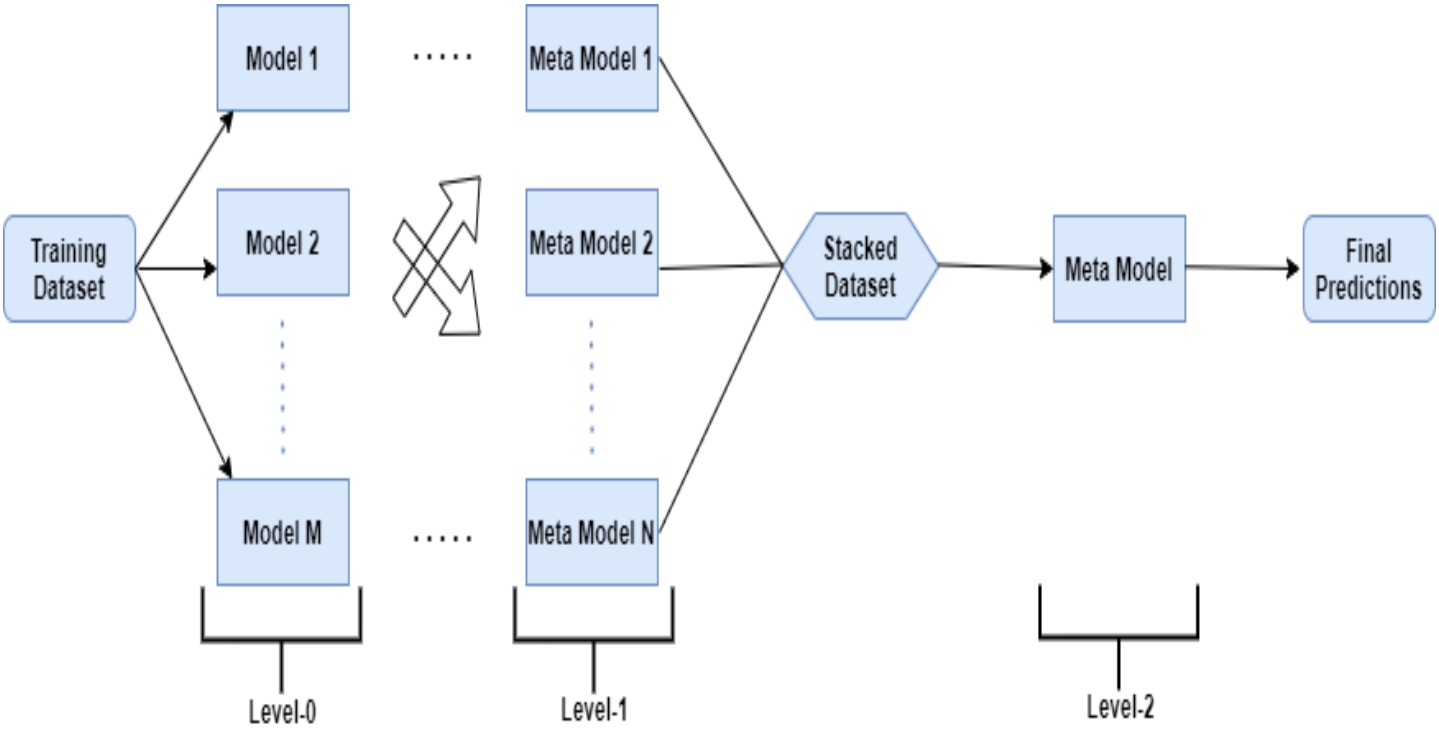
The level-0 models can be either a diverse range of algorithms or the same algorithm (most often they are diverse). The level-1 meta-model is most often a simple model, like Linear Regression for regression problems and Logistic Regression for Classification problems.

The stacking method can reduce the bias or variance based on the algorithms used in level-0.

There are many libraries for Stacking, like - StackingClassifier, StackingRegressor, make_classification, make_regression, ML Ensemble, and H20.

Multi-level Stacking

Multi-levels Stacking is an extension of stacking where stacking is applied on multiple layers.



For example in a 3-level stacking, Level-0 is the same where a diverse range of base learners are trained using k-fold cross-validation. In level-1, instead of a single meta-model, N such meta-models are used. In level-2, the final meta-model is used that takes the predictions from N meta-models of level-1.